



Finals

Introduction to Data Science (National University of Singapore)



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NATIONAL UNIVERSITY OF SINGAPORE

DSA1101 INTRODUCTION TO DATA SCIENCE

(Semester 2 : AY 2025/2026)

Time Allowed: 2 Hours

INSTRUCTIONS TO STUDENTS

1. Please indicate only your student number on your answer file (R code file).
Do not indicate your name.
2. This exam paper contains **FOUR (4)** problems and comprises **EIGHT (8)** pages (including the cover page).
3. This is an OPEN BOOK BLOCKED INTERNET exam. You may refer to your lecture notes, tutorials, textbooks or any notes that you have made, but you are not allowed to perform any online search.
4. The use of offline Large Language Models (LLMs) is prohibited for this exam.
5. All calculators in the List of Approved Calculators are allowed.
6. Students are required to answer ALL questions. Total mark is 100.
7. Student **MUST label systematically the answer for each question in the answer code.**
8. During the exam, you are not allowed to communicate with any person other than the invigilators.
9. At the end of the exam, **copy and paste your R code into the Exemplify text box for grading.**
10. Save an exact copy of your R code as an .R file, and upload it to Canvas/DSA1101/Assignments/Final Exam Submission immediately after the exam. Do not modify the code in your submission file. Any discrepancies will result in the question not being marked.

1. (20 points) Please type out your answers as comments in your answer file.
No explanation is needed.

- (a) The coefficient of determination of a multiple linear regression model, R^2 , will decrease when insignificant predictors are added to the model. True or False?
- (b) As long as the inputs are in the same unit, there is no need to standardize the inputs before feeding them into a KNN model for classification. True or False?
- (c) The coefficients in both linear and logistic regression models are estimated by minimizing the sum of squared errors. True or False?
- (d) In association rules, *Leverage* cannot be negative. True or False?
- (e) Data scientists should consider potential malicious uses of their models as part of ethical responsibility. True or False?
- (f) Suppose we have a student feedback data set about a learning course which has 10 inputs and a binary outcome variable Y that takes values 0 and 1. All of the inputs are continuous variables, except for two:

x_1 : The chapter in the course that student chose to write an essay for Assignment (takes values 1, 2, 3, 4, 5, 6, 7);

x_2 : Student's general feedback score about the course (takes values 1, 2, 3, 4, 5, from 1 = bad experience to 5 = great experience).

Which one of the following statements is true?

- A. KNN classification can be performed with all the 10 inputs.
 - B. KNN classification can be performed with the 8 continuous inputs only.
 - C. KNN classification can be performed with all the continuous inputs and x_1 only.
 - D. KNN classification can be performed with all the continuous inputs and x_2 only.
- (g) Which one of the following statements is true about the crucial assumption underlying the Naïve Bayes classifier?
- A. All the inputs must be categorical.
 - B. All the inputs are associated with each other.
 - C. The association of each input and the response is independent of each other.
 - D. All A, B and C are true.

- (h) Let `test.y` denote the vector of true labels in the test data. Consider the command below in R.

```
pred = knn(train = train.x, test = test.x, cl = train.y, k = 5, prob = TRUE)
```

What will the code below produce?

```
table(pred, test.y)
```

- A. The confusion matrix.
 - B. The table of predicted labels on the right column and true labels on the left column.
 - C. An error, this code will not run.
 - D. The predicted probabilities of the winning labels for the test set.
- (i) If a classification tree using Information Gain is allowed to grow without any stopping criterion, then which of the following statements would be true for the fitted tree?
- A. The number of internal nodes will be equal to the training sample size.
 - B. The number of leave nodes (leaves) will be equal to the training sample size.
 - C. Both A and B are true
- (j) In association rules, consider the rule $C \rightarrow D$. Which of the following statements is true?
- A. A low positive leverage implies that C and D occur together less often than expected by chance.
 - B. Confidence measures how often C and D occur together in the data set.
 - C. Lift greater than 1 indicates that C and D occur together more often than expected by chance.
 - D. Support measures how often D occurs given that C occurs.

2. (15 points) Consider a data set on the information of passengers on the Titanic. The description of variables in the data set is given below.

Variable	Description
Survived	survival status: 1 = Yes, 0 = No
Class	passenger's ticket class: 1st, 2nd, and 3rd
Sex	gender of the passenger: Male, Female
Age	age of the passenger: Child, Adult

A model, named model M, is fitted to the data set to predict the survival status of passengers. Summary output of model M is given in Figure 1 next page. Answer the questions below using Figure 1. **No explanation is needed.**

```
Call:
glm(formula = Survived ~ ., family = binomial, data = data)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.0063	0.1693	11.851	< 2e-16 ***
Class2nd	-1.0106	0.1949	-5.184	2.17e-07 ***
Class3rd	-1.7664	0.1707	-10.347	< 2e-16 ***
SexMale	-2.3695	0.1453	-16.313	< 2e-16 ***
AgeChild	1.0556	0.2427	4.350	1.36e-05 ***

Figure 1: Summary output of model M

- (a) Model M is a linear regression model. True or False?
- (b) Model M's prediction on the probability of survival for an adult passenger A who is a male and having ticket in 3rd class is
- A. 0.119
 - B. -2.13
 - C. 0.106
 - D. 0.255
- (c) Passenger B is a female adult who has 3rd class ticket. The odds ratio of survival of passenger B compared to that of passenger A is
- A. 2.3695
 - B. 10.692
 - C. -2.3695
 - D. 0.453
- (d) Given two passengers of the same age and gender, the odds ratio of survival of a passenger with ticket in the 2nd class compared to that of a passenger with ticket in the 3rd class is
- A. 0.7558
 - B. -0.7558
 - C. 0.6804
 - D. 2.129

3. (15 points) Tardive dyskinesia (TD) is a movement disorder that may develop in patients on long-term neuroleptic treatment. It has previously been established that factors like age, gender and duration of exposure to neuroleptics are risk factors. A research study is conducted to investigate the effects of these factors. The table below describes some variables in this study.

Variable	Description
age	age of subject at time of study (year)
sex	1 = female; 2 = male
durill	duration of time that a subject has been taking the neuroleptic (year)
cpz	daily dose of chlorpromazine (mg) - a neuroleptic
td	status of tardive dyskinesia: 1 = affected; 3 = unaffected

A step of model fitting is given by an R command in Figure 2. The fitted tree obtained by the command is given in Figure 3.

For all questions in this problem, please report the values as comments in the answer code. **No explanation is needed.**

```
dt = rpart(td ~ age + sex + durill + cpz,
           method="class",
           data=data,
           control=rpart.control(maxdepth = 5))
```

Figure 2: R command to fit a decision tree for Tardive data set

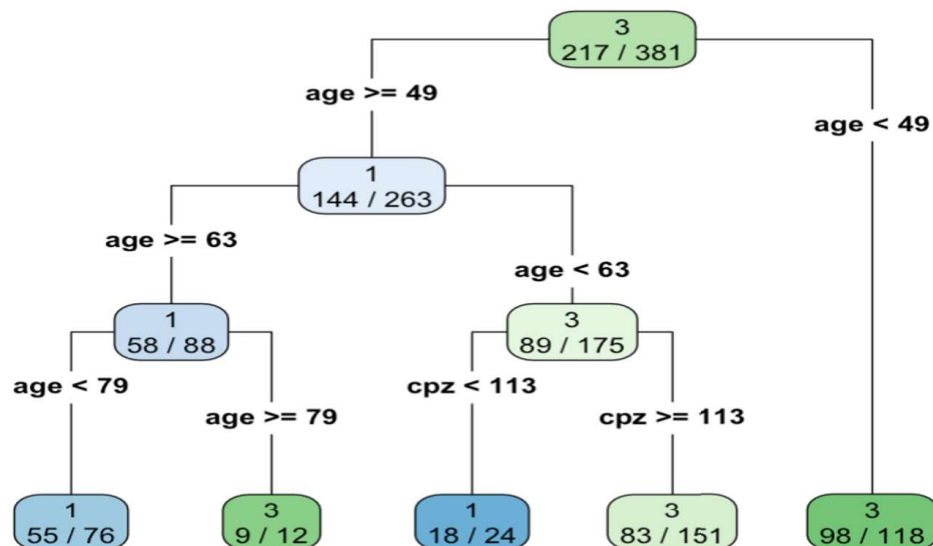


Figure 3: The fitted tree

(a) Report the sample size.

Report the proportion of the sample that is affected by TD (fraction or 2 d.p.).

(b) The fitted tree doesn't use all the input variables that were proposed by the model fitting step. True or False?

(c) Report the most important factor in developing TD among patients on long-term neuroleptic treatment.

(d) The fitted tree is obtained by using "Information Gain" for branch split. True or False?

(e) Patient A with **age** = 48, **sex** = female, **durill** = 5 and **cpz** = 100. Report the chance of being affected by TD for patient A (fraction or 3 d.p.).

(f) Patient B with **age** = 60, **sex** = male, **durill** = 10 and **cpz** = 120. Report the chance of being unaffected by TD for patient B (fraction or 2 d.p.).

(g) For the given fitted tree, if a patient is of 65 years old, then all the information about gender, duration of time that patient was on neuroleptic treatment and cpz dosage is not important to predict if patient is affected by TD. True or False?

4. (50 points) In R, run three lines of code below.

```
library(MASS)
data(Pima.tr)
df = Pima.tr
```

The names given in bold below **MUST** be used in your R code.

df is a data frame of 8 variables that studies the risk factors that may associate with diabetes among women of Pima Indian heritage and living near Phoenix, Arizona.

df will be used for this problem where the description of the variables is given below.

Variable	Description
npreg	number of pregnancies
glu	plasma glucose concentration in an oral glucose tolerance test (mg/dL)
bp	diastolic blood pressure (mm Hg)
skin	triceps skin fold thickness (mm)
bmi	body mass index
ped	diabetes pedigree functions – a score to summarize a person's family history of diabetes
age	age (year)
type	diabetes status, Yes or No

For the questions below, do not split the data set given into train set and test set.

Part I: Logistic Regression Models (15 points)

- (a) Write code to create a new column in **df** named **y**, where **y** equals 1 if **type** is Yes, and 0 otherwise.
Report the proportion of observations where **y** equals 1 (2 d.p.).
- (b) Write code to fit a logistic regression model using all the input features, to be named as **LR**, for predicting the probability of having diabetes.
Using a significance level of 0.1, report the name(s) of any insignificant input features in model **LR**.
- (c) Using **LR**, compute and report the probability of having diabetes for two person, A and B, whose information is listed below (4 d.p.).
A: npreg = 1, glu = 100, bp = 64, skin = 21, bmi = 27, ped = 0.25, age = 23;
B: npreg = 3, glu = 124, bp = 72, skin = 30, bmi = 32, ped = 0.46, age = 48.
- (d) With the predicted probabilities for two persons, A and B, obtained from part (c), how would you predict the diabetes status for each of them? Explain your prediction.
- (e) Write a code to plot the ROC curve of the model **LR** in red color based on its predictions for the data given. Derive and report the value of AUC (2 d.p.).

Part II: Decision Trees (15 points)

All the decision tree models below are using Information Gain for branch split.

Response **y** is used.

- (f) Using only **minsplit** parameter with **minsplit** = 20 to specify the complexity of the tree, write code to fit a decision tree, named **DT**, with all the input features given in the data set.
Report the names of the inputs that appear in the fitted tree. Among those inputs, which one is the most important?
- (g) From the plot of the tree, obtain and report the predicted probability of having diabetes for the two persons, A and B in part (c) (fraction or 4 d.p.).
- (h) Write code to predict the status of diabetes of people in **df**. The vector of the status is named as **DT.pred**.
Calculate and report Type I and Type II error rates for the predicted status (4 d.p.).

- (i) Write code to predict the probability of having diabetes for people in **df** where the vector of probabilities is named as **DT.prob**.

Write a code to plot the ROC curve of the model **DT** in blue color based on the predicted probabilities **DT.prob**.

Derive and report the value of AUC (3 d.p.).

Part III: Naive Bayes Classifier (20 points)

- (j) We now fit a naive Bayes classifier using data frame **df** with all the inputs given. Write code to fit the classifier, to be named as **NB**.

Write code to get the predicted probability of having diabetes for two person, A and B, mentioned in part (c). Report the probabilities (4 d.p.).

- (k) Write code to predict the probability of having diabetes for people in **df** where the vector of probabilities is named as **NB.prob**.

Report the number of people that has the predicted probability of having disease larger than 0.5.

- (l) Let **alpha** denote a vector of values for threshold in the range from 0.2 to 0.8 inclusive and rounded to 1 decimal place. For each value of threshold in **alpha**, write code to

- (i) predict the status of having diabetes for people in **df**,
- (ii) calculate Type I and Type II error rates, and store those values in vectors named **type1.nb** and **type2.nb**, respectively.

Copy and paste the matrix of three columns, **alpha**, **type1.nb** and **type2.nb**, into the answer file as comments.

- (m) When the threshold values in **alpha** increases from 0.2 to 0.8, what is the trend for the values of Type I error rate in **type1.nb**, and what is the trend for the values of Type II error rate in **type2.nb**?

Can you explain the logic for the observed trends to happen?

- (n) Write code to plot the ROC curve of the classifier **NB** in black color based on its prediction for **df**. Derive and report the value of AUC (3 d.p.).

- (o) In the plot in part (n), write code to add the ROC curve of **LR** in red color and ROC curve of **DT** in blue color.

Add a legend for the plot where each curve has its AUC value shown in the plot. Which curve has the highest AUC value?

–END OF PAPER–